

Thatipally Aadithhya

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Education

Indian Institute of Information Technology, Nagpur

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.26 / 10.0

Nagpur, India

Aug 2023 – Jun 2027

Experience

Futures First

Jul 2026 – Present

Software Development Engineer Intern, Quant & Algo

Hyderabad, India

- Cut order-placement latency from **2s to <10ms** by re-engineering backend order-management state-handling logic in **C#/.NET**, improving throughput under high-concurrency conditions.
- Engineered low-latency trading pipelines processing **10,000+ trades daily** with millisecond-level execution on live production trading workflows.
- Implemented **10+ risk and compliance controls** (self-trade prevention, price limits, margin checks) and built a **React**-based real-time dashboard for order status and latency monitoring.

AeroAgro AI (IIT Hyderabad)

Apr 2025 – Jul 2025

Project Intern

Hyderabad, India

- Developed a **Node.js** drone-image upload pipeline with input validation, structured error handling, and **Cloudinary CDN** integration, reducing failed transfers by **30%** and enabling image retrieval in under **2 seconds**.
- Collaborated cross-functionally with a **5+ member** engineering and hardware team, translating on-drone requirements into backend features to ensure seamless hardware-software integration.

Projects

QuickLearn | Next.js, PostgreSQL, Gemini API

GitHub | Live Demo

- Designed an end-to-end adaptive-learning pipeline integrating the **Google Gemini LLM** to auto-generate structured learning modules, chapter breakdowns, and personalized roadmaps from a single topic prompt in under **10 seconds**, reducing manual course-creation time by **80%**.
- Architected a **PostgreSQL**-backed progress-tracking pipeline with **YouTube API** integration for curated video content, increasing user engagement by **25%**.

LiveDocs - Live Google Docs | TypeScript, Node.js, WebSockets, PostgreSQL

GitHub

- Built a production-grade, distributed, real-time collaborative document editor supporting **10+ concurrent users** with sub-**50ms** sync latency, implementing a custom **CRDT algorithm** to eliminate merge conflicts across sessions.
- Architected secure multi-user access with **JWT**-authenticated **WebSocket** connections, room-based isolation, and granular per-user permissions across a distributed, multi-tiered system, ensuring zero unauthorized data exposure.

Astra | C, Flex (Lex), Bison (Yacc), GCC

GitHub

- Devised a fully functional compiler from scratch, building a **Flex** lexer for lexical analysis and a **Bison** context-free-grammar parser for syntax parsing of a custom Hindi-inspired programming language.
- Designed an **Abstract Syntax Tree (AST)** and devised a **O(n)** single-pass code generation across **5+ language constructs** (variables, functions, conditionals, loops, I/O) via optimized **DFS traversal**, producing GCC-runnable binaries.

Technical Skills

Languages: C++, Java C, C#, Go, JavaScript, TypeScript

Backend & Systems: .NET, Node.js, Express.js, REST APIs, WebSockets, Multithreading, Distributed Systems

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Databases: PostgreSQL, MySQL, MongoDB, Redis

Developer Tools: Git, GitHub, Linux, Postman, Vercel

Coursework: System Design, Operating Systems, DBMS, Computer Networks, Object-Oriented Programming

Achievements & Certifications

- **Competitive Programming:** LeetCode Rating **1800+**; solved **400+** problems across data structures and algorithms.
- **Contest Rankings:** Global Rank **769** in CodeChef Contest 234C; solved **200+** problems.
- **Hackathons:** Reached the national semi-finals of Flipkart GRiD 8.0, ranking among the top **1,000** out of **1,65,000+** registrations nationwide.

Leadership Experience

Indian Institute of Information Technology, Nagpur

Feb 2025 – Jul 2025

Student Placement Cell Coordinator

Nagpur, India

- Managed end-to-end placement operations, including interview scheduling, candidate data management, and company coordination, for **50+ companies** and **300+ students**.